

AP CALCULUS AB – UNIT 5

Analytical Applications of Differentiation

Expanded Complete-Coverage Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Verify hypotheses and apply Rolle's Theorem and the Mean Value Theorem.
- Use first-derivative information to determine intervals of increase/decrease and relative extrema.
- Use second-derivative information to determine concavity and points of inflection.
- Connect the graphs of a function and its derivatives and justify global conclusions.
- Construct and solve optimization models with domain and endpoint analysis.
- Review every idea, theorem hypothesis, representation, procedure, and common misconception appearing in the website lessons.

Complete Lesson Coverage Audit

Every website lesson is given its own mastery section below. Each section contains the complete lesson objectives, strategic reminders, a solved model, and foundation-to-challenge practice. The final synthesis sections mix lessons so that students must select procedures rather than follow a stated template.

Complete Website Lesson Coverage Map

Lesson	Website title	Ideas explicitly reviewed
5.1	Mean Value Theorem	Verify continuity and differentiability hypotheses for the Mean Value Theorem.; Find or guarantee points where instantaneous and average rates agree.; Apply Rolle's Theorem as a special case.; Distinguish existence conclusions from exact location and from the Intermediate Value Theorem.
5.2-5.5 I	Derivative Graph Analysis - Part 1	Find critical numbers and intervals of increase and decrease.; Use the First Derivative Test to classify local extrema.; Apply the Extreme Value Theorem and closed-interval method for absolute extrema.; Connect the sign of f' with the shape of f .
5.2-5.5 II	Derivative Graph Analysis - Part 2: Challenge Questions	Determine intervals of concavity and locate points of inflection.; Use the Second Derivative Test when its hypotheses are satisfied.; Relate the graphs and signs of f , f' , and f'' .; Synthesize first- and second-derivative information in complete curve analysis.
5.10	Optimization Problems	Identify an objective quantity and a constraint relating variables.; Rewrite the objective as a one-variable function on a meaningful domain.; Find and justify an absolute maximum or minimum using critical points and endpoints.; Interpret optimized dimensions and values with correct units.

Unit Knowledge, Definitions, and Theorem Library

Key Knowledge, Formulas, and Theorems

- MVT requires continuity on $[a, b]$ and differentiability on (a, b) , then guarantees $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some $c \in (a, b)$.
- Rolle's Theorem is the MVT case $f(a) = f(b)$ and guarantees at least one interior horizontal tangent.
- A theorem can be inapplicable even if its conclusion happens to occur; always verify the hypotheses first.
- Critical numbers lie in the domain where $f' = 0$ or f' does not exist. A sign change in f' classifies a local extremum.
- EVT guarantees absolute extrema for a continuous function on a closed interval. The closed-interval method compares endpoints and all interior critical numbers.
- $f' > 0$ means increasing and $f' < 0$ means decreasing. When reading a graph of f' , use its sign for monotonicity of f .
- $f'' > 0$ means concave up and $f'' < 0$ means concave down. An inflection point requires an actual concavity change, not merely $f'' = 0$.
- When reading a graph of f' , the increase/decrease of f' gives concavity of f ; extrema of f' correspond to candidates where $f'' = 0$.
- The Second Derivative Test classifies a critical point only when $f'(c) = 0$ and $f''(c)$ has a nonzero sign; $f''(c) = 0$ is inconclusive.
- Optimization requires an objective, a constraint, a one-variable model, a feasible domain, critical and endpoint candidates, a global justification, and a contextual final answer.

Strategy and Precision Checklist

1. Name the representation and the requested quantity before calculating.
2. State all hypotheses before using a theorem.
3. Preserve exact values in non-calculator work and use radians for trigonometric calculus.
4. A graph or table supports an estimate; an analytic proof requires a definition, theorem, or valid algebraic argument.
5. Contextual conclusions must include the instant, sign, units, and meaning.
6. Candidate conditions such as a zero derivative, zero second derivative, or zero denominator require sign/domain verification before a conclusion.

Lesson 5.1: Mean Value Theorem**Complete objectives**

- Verify continuity and differentiability hypotheses for the Mean Value Theorem.
- Find or guarantee points where instantaneous and average rates agree.
- Apply Rolle’s Theorem as a special case.
- Distinguish existence conclusions from exact location and from the Intermediate Value Theorem.

Strategic reminders

1. State both hypotheses on their correct intervals before writing the conclusion.
2. Compute the endpoint secant slope, then solve $f'(c)$ equal to that slope and keep only solutions in (a, b) .
3. Do not use endpoint values alone to claim differentiability.
4. A theorem may be inapplicable even if a point satisfying the conclusion happens to exist.
5. In context, interpret the average and instantaneous rates with units.

Solved Example: Polynomial MVT

Verify the MVT for $f(x) = x^2 - 2x$ on $[0, 3]$ and find c .

Answer and Reasoning

A polynomial is continuous and differentiable. The secant slope is $(3 - 0)/3 = 1$. Since $f'(x) = 2x - 2$, solve $2c - 2 = 1$ to get $c = 3/2$.

Lesson Mastery Practice

1. State the two MVT hypotheses.

Foundation

2. Apply the MVT to $f(x) = 1/x$ on $[1, 4]$ and find c .

Core

3. Find all MVT points for $f(x) = x^3 - 3x$ on $[-2, 2]$.

Advanced

4. Use MVT to prove $|\sin b - \sin a| \leq |b - a|$.

Challenge

AP-Style Reasoning Check

MVT requires a function to be

- (A) continuous on (a, b) only (B) differentiable on $[a, b]$
(C) continuous on $[a, b]$ and differentiable on (a, b) (D) increasing on $[a, b]$

Lesson 5.2-5.5 I: Derivative Graph Analysis - Part 1**Complete objectives**

- Find critical numbers and intervals of increase and decrease.
- Use the First Derivative Test to classify local extrema.
- Apply the Extreme Value Theorem and closed-interval method for absolute extrema.
- Connect the sign of f' with the shape of f .

Strategic reminders

1. Find the domain first, then solve $f' = 0$ and include points in the domain where f' is undefined.
2. Build a sign chart for f' using test inputs or factor signs.
3. State extrema as points or values as requested, and distinguish local from absolute.
4. For an absolute-extrema problem on $[a, b]$, endpoints are mandatory candidates even though derivatives are not required there.
5. When given the graph of f' , local extrema of f occur at sign changes, not merely at zeros of f' .

Solved Example: First derivative sign chart

For $f(x) = x^3 - 3x^2 - 9x + 1$, find intervals of increase/decrease and local extrema.

Answer and Reasoning

$f' = 3(x - 3)(x + 1)$. Increasing on $(-\infty, -1) \cup (3, \infty)$, decreasing on $(-1, 3)$. Local max at $x = -1$ with $f(-1) = 6$; local min at $x = 3$ with $f(3) = -26$.

Lesson Mastery Practice

5. Find critical numbers of $f(x) = x^2 - 6x + 1$.

Foundation

6. For $f' = x(x - 4)$, determine intervals where f increases.

Core

7. For $f(x) = x + 4/x$ on $(0, \infty)$, find the absolute minimum.

Advanced

8. Construct a differentiable function with a critical point at 0 that is neither maximum nor minimum.

Challenge

AP-Style Reasoning Check

A critical number must

- (A) make $f' = 0$ only
- (B) be in the domain of f and make $f' = 0$ or undefined
- (C) be an endpoint
- (D) be a local extremum

Lesson 5.2-5.5 II: Derivative Graph Analysis - Part 2: Challenge Questions**Complete objectives**

- Determine intervals of concavity and locate points of inflection.
- Use the Second Derivative Test when its hypotheses are satisfied.
- Relate the graphs and signs of f , f' , and f'' .
- Synthesize first- and second-derivative information in complete curve analysis.

Strategic reminders

1. Solve $f'' = 0$ and include points where f'' is undefined, then test concavity on the resulting intervals.
2. Verify an actual sign change in concavity before naming an inflection point.
3. When reading f' , distinguish its sign (monotonicity of f) from its slope (concavity of f).
4. The Second Derivative Test applies only at a critical point with a suitable second derivative sign.
5. A complete AP explanation links the derivative evidence to a precise conclusion about the original function.

Solved Example: Concavity

For $f(x) = x^4 - 4x^3$, determine concavity and inflection points.

Answer and Reasoning

$f'' = 12x(x - 2)$. Concave up on $(-\infty, 0) \cup (2, \infty)$ and down on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$, with points $(0, 0)$ and $(2, -16)$.

Lesson Mastery Practice

9. If $f'' > 0$ on an interval, describe concavity.

Foundation

10. For $f(x) = x^3 - 6x$, find the inflection point.

Core

11. For $f(x) = x^4 - 8x^2$, find local extrema and inflection points.

Advanced

12. Construct a function with $f''(0) = 0$ but no inflection at 0.

Challenge

AP-Style Reasoning Check

An inflection point requires

- | | |
|---------------------------|----------------------|
| (A) $f'' = 0$ only | (B) a local extremum |
| (C) a change in concavity | (D) $f' = 0$ |

Lesson 5.10: Optimization Problems**Complete objectives**

- Identify an objective quantity and a constraint relating variables.
- Rewrite the objective as a one-variable function on a meaningful domain.
- Find and justify an absolute maximum or minimum using critical points and endpoints.
- Interpret optimized dimensions and values with correct units.

Strategic reminders

1. Draw and label a model, then state the objective and constraint separately.
2. Use the constraint to eliminate variables until the objective depends on one variable.
3. Write the physically meaningful domain before differentiating.
4. Find critical points and compare all feasible candidates; do not stop after solving $F' = 0$.
5. Answer the original question, not merely the value of the chosen variable.

Solved Example: Maximum rectangle

A rectangle has perimeter 40 m. Find dimensions of maximum area.

Answer and Reasoning

$2x + 2y = 40$, so $y = 20 - x$. $A = x(20 - x)$ on $0 < x < 20$. $A' = 20 - 2x = 0$ gives $x = 10$, $y = 10$; the maximum area is 100 m^2 .

Lesson Mastery Practice

13. For a rectangle with fixed perimeter P , express y in terms of x .

Foundation

14. A rectangle has area 96 m^2 . Express perimeter as a function of width x .

Core

15. Find the maximum area in the river-fencing problem.

Advanced

16. A rectangle is inscribed under $y = 12 - x^2$ above the x -axis, symmetric about the y -axis. Maximize area.

Challenge

AP-Style Reasoning Check

The objective function is

- | | |
|-----------------------------|--|
| (A) the fixed equation only | (B) the quantity to maximize or minimize |
| (C) the derivative | (D) the domain |

Unit 5 Difficult and Challenge Synthesis

1. A function is continuous on $[0, 4]$, differentiable on $(0, 4)$, and satisfies $f(4) - f(0) = 12$. If $f'(x) \leq 2$ for every $x \in (0, 4)$, explain the contradiction. **Hard**

2. Let $f'(x) = x(x - 2)^2(x + 1)$. Determine all intervals of increase/decrease and classify every local extremum of f .

Difficult

3. Suppose the graph of f' is positive and decreasing on $(0, 2)$, negative and decreasing on $(2, 5)$, and negative and increasing on $(5, 7)$. Describe monotonicity, local extrema, and concavity of f . **Hard**

4. Construct a differentiable function with $f'(0) = 0$ and $f''(0) = 0$ but no local extremum and an inflection point at 0.

Hard

5. Find the absolute extrema of $f(x) = x^{2/3}(x-2)$ on $[-1, 4]$, including nondifferentiable critical numbers.

Challenge

6. An open-top box is made from a 24×18 rectangle by cutting squares of side x . Find the globally maximizing cut size and justify it.

Challenge

Cumulative AP-Style Checkpoint

1. Which hypotheses are sufficient for the Mean Value Theorem on $[a, b]$?

- (A) Continuity on (a, b) only (B) Equal endpoint values only
(C) Continuity on $[a, b]$ and differentiability on (a, b) (D) Differentiability only at the endpoints

2. If f' changes from positive to negative at $x = c$, then f has

- (A) a local minimum at c (B) an inflection point at c
(C) no possible conclusion (D) a local maximum at c

3. Which condition is required for an inflection point?

- (A) A local maximum (B) A change in concavity
(C) $f' = 0$ (D) $f'' = 0$

4. In an optimization problem, after expressing the objective as a one-variable function, the next essential step is to

- (A) ignore endpoints (B) differentiate the constraint instead of the objective
(C) state the feasible domain before finding and comparing candidates (D) assume every critical point is the global optimum

5. If $|f'(x)| \leq 4$ on $(1, 6)$, then the greatest possible value of $|f(6) - f(1)|$ is bounded by

- (A) 24 (B) 20
(C) 4 (D) 5

6. Which input is a critical number of f ?

- (A) Every endpoint, whether or not in the domain (B) An input in the domain where $f' = 0$ or f' does not exist
(C) Every input where f does not exist (D) Only inputs where $f' = 0$

7. If f' is positive and decreasing on an interval, then f is

- (A) decreasing and concave down (B) decreasing and concave up
(C) increasing and concave down (D) increasing and concave up

8. What dimensions maximize the area of a rectangle with perimeter 60?

- (A) 15 by 16.0 (B) 15 by 15
(C) 30 by 30 (D) 1 by 29

9. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

- (A) 26 (B) $\frac{5\sqrt{3}}{3}$
(C) $\frac{5}{3}$ (D) 5

10. If $f'(x) = (x + 2)(x - 4)$, on which interval is f decreasing?

- (A) $(-\infty, -2) \cup (4, \infty)$ (B) $(-2, 4)$
(C) $(4, \infty)$ (D) $(-\infty, -2)$

11. If $f'(c) = 0$ and $f''(c) = 0$, the Second Derivative Test is

- (A) inconclusive (B) a proof of a local maximum
(C) a proof of a local minimum (D) a proof of an inflection point

12. What dimensions maximize the area of a rectangle with perimeter 52?

- (A) 13 by 13 (B) 1 by 25
(C) 13 by 14.0 (D) 26 by 26

13. For $f(x) = x^3$ on $[0, 3]$, find a value c guaranteed by the Mean Value Theorem.

- (A) $\sqrt{3}$ (B) 3
(C) 1 (D) 10

14. Suppose f' changes from positive to negative at $x = 3$. What occurs there?

- (A) A vertical tangent of f (B) An inflection point of f
(C) A relative maximum of f (D) A relative minimum of f

15. Find the inflection point x -coordinate of $f(x) = x^3 - 3x$.

- (A) -1 (B) 2
(C) 0 (D) 1

16. A rectangle has perimeter 20. If its width is x , which area model uses one variable?

- (A) $A(x) = x^2(10 - x)$ (B) $A(x) = 2x(10 - x)$
(C) $A(x) = x(10 - x)$ (D) $A(x) = x(20 - x)$

17. For $f(x) = x^3$ on $[0, 5]$, find a value c guaranteed by the Mean Value Theorem.

- (A) 5 (B) 26
(C) $\frac{5}{3}$ (D) $\frac{5\sqrt{3}}{3}$

18. Which set must be checked to find an absolute maximum of a continuous function on a closed interval?

- (A) Only points where $f'(x) = 0$ (B) Endpoints and interior critical points
(C) All points where $f''(x) = 0$ (D) Only endpoints

19. At a point where $f'(c) = 0$ and $f''(c) < 0$, the second derivative test concludes

- (A) no conclusion is possible (B) f has a local maximum at c
(C) f has a local minimum at c (D) f has an inflection point at c

20. A rectangle has perimeter 48. If its width is x , which area model uses one variable?

- (A) $A(x) = x(24 - x)$ (B) $A(x) = x(48 - x)$
(C) $A(x) = x^2(24 - x)$ (D) $A(x) = 2x(24 - x)$

Cumulative Free-Response Synthesis

FRQ1 – Derivative graph synthesis

The derivative f' is continuous on $[-4, 5]$ and has zeros at $x = -2, 1, 4$. It is positive on $(-4, -2)$ and $(1, 5)$, negative on $(-2, 1)$, increasing on $(-4, -1)$ and $(2, 5)$, and decreasing on $(-1, 2)$.

(a) Determine intervals of increase/decrease of f and classify all local extrema. [4 points]

(b) Determine intervals of concavity and all inflection inputs. [3 points]

(c) At which listed zero of f' is the Second Derivative Test inconclusive from the information given? [2 points]

FRQ2 – Optimization with a calculator

An open-top box is made from a $24 \text{ cm} \times 18 \text{ cm}$ rectangle by cutting squares of side x from each corner.

(a) Write the volume function and its feasible domain. [2 points]

(b) Find the critical points and identify the feasible one. [3 points]

(c) Justify that the feasible critical point produces the absolute maximum and report the maximum volume. [3 points]

FRQ3 – Mean Value Theorem and derivative bounds

A differentiable function f satisfies $f(0) = -2$ and $f(5) = 13$.

- (a) State all hypotheses needed to guarantee a point c with $f'(c) = 3$. [3 points]
- (b) Explain why the additional claim $f'(x) \leq 2$ for all $x \in (0, 5)$ is impossible under those hypotheses. [2 points]
- (c) Does MVT guarantee that this input is unique? Explain. [1 points]

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